

LABORATORY REPORT


Name	:Ms. BASANTI DEVI 10078040	Sex/Age	: Female/50 Years	Case ID	:60400127778
	OQG2604170493				
Ref By	:	Dis.Loc.	:	Pt ID	:
Bill. Loc.	:Oncquest Laboratories Ltd.			Pt. Loc.	:
Registration Date & Time	: 17-Apr-2026 16:42	Sample Type	: Whole Blood EDTA	Ph #	: 7830215230
Sample Date & Time	: 17-Apr-2026 16:42	Sample Coll.By	:	Ref Id	:
Report Date & Time	: 09-May-2026 14:08	Acc. Remarks	:	Ref Id 2	:

BCR::ABL1 KINASE DOMAIN MUTATION ANALYSIS (IRMA) BY NEXT GENERATION SEQUENCING

Test Description:	Chronic myeloid leukemia (CML) is characterized by the presence of the Philadelphia chromosome, the product of t(9;22)(q34;q11) translocation. This translocation results in BCR-ABL1 fusion protein with constitutive ABL1 tyrosine kinase activity. The kinase inhibitor imatinib (STI571, Gleevec®) inhibits ABL1 kinase activity and is now the standard of care for early phase CML. Prolonged treatment with imatinib can lead to drug resistance, especially in patients with advanced disease. A large portion of resistant patients have acquired point mutations in the ABL1 kinase domain that renders the kinase resistant to the drug. Sites of point mutations in ABL1 associated with imatinib resistance span the entire kinase domain but often cluster in important hotspots.(1,2) Total RNA was extracted and nested reverse transcription PCR was performed to detect the BCR-ABL1 transcript and ABL1 kinase domain (KD) region was thereafter sequenced by Next Generation Sequencing for detection of mutant alleles.
Methodology:	Next Generation Sequencing (Massively Parallel Sequencing)
Sample:	EDTA Blood

Test results and Interpretation:	<u>PATHOGENIC MUTATION DETECTED</u>
Mutation Nomenclature:	L248V
Coverage:	>500x

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Interpretation:
L248V [VAF: 99.81%]

 HGVS genomic change: **9:g.133738342C>G**

 HGVS protein change: **ENSP00000323315:p.Leu248Val**

 HGVS coding DNA change: **ENST00000318560.5:c.742C>G**

 Mutation type: **Pathogenic**

The detection of p. Leu248Val (p.L248V) mutation has been described in association with resistance to Imatinib (PMID: 21562040)

Final conclusion:

The patient is found to have L248V mutation.

Sr. No.	Mutations covered	Mutation Result	Sensitivity to Imatinib	Mean fold Increase in IC50 (Imatinib)2	Sr. No.	Mutations covered	Mutation Result	Sensitivity to Imatinib	Mean fold Increase in IC50 (Imatinib)2
1	T315I	Negative	R	50.8	19	E275K	Negative	NA	
2	E255V	Negative	R	44.3	20	F359C	Negative	NA	4.3
3	Y253H	Negative	R	36.3	21	E450K	Negative	NA	3.9
4	E255K	Negative	R	30.7	22	M351T	Negative	S	3.8
5	Y253F	Negative	R	25.2	23	M244V	Negative	I	3.8
6	E373G	Negative	NA	20.6	24	E355G	Negative	I	3.4
7	L248V	Positive	NA	19.9	25	V379I	Negative	I	3.1
8	G321E	Negative	NA	18.1	26	E355A	Negative	NA	3.0

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9	E279K	Negative	NA	16.5	27	M388L	Negative	NA	2.7
10	G250E	Negative	I	11.7	28	T315A	Negative	S	2.4
11	D276G	Negative	NA	11.2	29	M237I	Negative	NA	2.4
12	F486S	Negative	NA	8.4	30	F311L	Negative	S	2.2
13	H396P	Negative	S	7.1	31	L387F	Negative	NA	2.0
14	H396R	Negative	I	6.7	32	V299L	Negative	S	1.7
15	F359V	Negative	I	5.8	33	F317V	Negative	S	1.4
16	F317L	Negative	I	4.8	34	G250A	Negative	NA	1.3
17	Q252H	Negative	I	4.8					
18	L387M	Negative	S	4.3					

SENSITIVE- S

RESISTANT- R

INTERMEDIATE REPSPONE- I

NOT APPLICABLE- NA

Novel Mutation	Nomenclature	Computationally Predicted Effect
-	-	-

REFERENCES:

- O'Hare T, Eide CA, Deininger MWN BCR-ABL kinase domain mutations, drug resistance and the road to a cure of chronic myeloid leukemia. Blood 2007; 110: 2242-9.*

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- Guidelines for mutation analysis of BCR/ABL kinase domain: Interpreting TKI-Resistance Mutations in CML Patients West Midlands Regional Genetics Laboratory March 2007; Revised December 2007.
- Soverini S, Hochhaus A, Nicolini FE, et al. BCR-ABL kinase domain mutation analysis in chronic myeloid leukemia patients treated with tyrosine kinase inhibitors: recommendations from an expert panel on behalf of European LeukemiaNet. Blood. 2011;118(5):1208-1215.
- von Bubnoff N, Schneller F, Peschel C, Duyster J. BCR-ABL gene mutations in relation to clinical resistance of Philadelphia-chromosome-positive leukaemia to STI571: a prospective study. Lancet. 2002;359(9305):487-491.
- De Benedittis, Caterina (2015) Next-Generation Sequencing-Based Mutations Scanning Strategy of the BCR-ABL Kinase Domain in Patients with Philadelphia-Chromosome Positive Leukemias Treated with Tyrosine Kinase Inhibitors.

Limitations:

- Please note that only the variants relevant to the patient's clinical presentation were reported. A report on all variants present is available upon request, please contact the laboratory.
- Predisposition to disease may be affected by non genetic factors.
- This test has not been approved by external accrediting bodies and has been validated according to internal laboratory criteria. Accreditation is not required for the reporting of these test results.
- For Accurate detection of mutation, at least 1000 BCR-ABL1 Fusion copies/ml should be present in the sample.
- In cases where the BCR ABL1 load is very low, cases may be released as 'No Mutation Detected' with a comment 'Repeated sequencing revealed no sequenceable Amplicons due to low transcript levels'
- All laboratory tests are associated with an error rate of ~1%. These could be due to sample mismatch, inappropriate labeling, processing or technological limitations. Please correlate with clinical features and other investigations for final conclusion and send a repeat sample for analysis if necessary.
- Rare false positive or negative results may occur due to inherent limitations of the technique involved.
- Large insertions and deletions may be missed.
- Mutations having VAF (variant allele frequency) less than 5% are not reported.
- For test performed on specimens received or collected from non-STMP locations, it is presumed that the specimen belongs to the patient named or identified as labeled on the container/test request and such verification has been carried out at the point of generation of the said specimen by the sender.

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